SPE-226845-MS

Integrated Analysis of Hydraulic Fracture, Microseismic & Pressure-Interference Test in a Dynamic Simulator to Optimize Well Spacing

Abdullah A. Almubarak and Adedayo D. Adebamiro, Saudi Aramco, Dhahran, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

It is challenging to simulate unconventional reservoirs efficiently and accurately because of uncertainty in fracture geometry, geomechanics, reservoir parameters, and a lack of holistic physics to integrate data from these three disciplines. We present a holistic approach to accurately predict the pad performance of hydraulically fractured wells. This involves extensive calibration of a frac model to Diagnostic Fracture Injection Test (DFIT) and microseismic data, which is validated by Rate Transient Analysis (RTA) and pressure interference test.

A single stage of a multistage fracture horizontal wells was calibrated with DFIT using a hydraulic fracture model. Simultaneously, the hydraulic fracture length (xf) was calibrated to align with microseismic events. This extensively calibrated stage was then replicated across other stages in the pad. Chow Pressure Group Analysis (CPG) was conducted on the four wells on the pad to measure the magnitude of the pressure interference. Results of the fracture model were passed into an in-house reservoir simulator for history matching.

This method serves as a way for frac and simulation engineers to create an accurate, calibrated model that can be used for frac, completion and well spacing optimization, technical and business decisions in areas of fields with similar petrophysics and geomechanical properties.

Introduction

Production forecasting, Estimated Ultimate Recovery, estimation of the hydrocarbon originally in place, well spacing, and how hydraulic fracture affects recovery from low permeability reservoirs are of great interest and concern to reservoir engineers in the gas and oil industry. Historically, this has been done using decline curve analysis, material balance, rate transient analysis, and reservoir simulation.

For a project to be successful, it is critical to know the well spacing, to characterize the hydraulic fracture geometry, to be able to predict how much oil and gas the well will produce over its life span, and to be able to test the impact of completion parameters and fracture design. In order to determine this, some oil companies carry out field spacing tests to determine the optimum well spacing scenario. While this is an effective solution, it takes time (2 to 3 years) to create enough groups of wells to study, it is expensive, and might not effectively capture all parameters that impact production. Other companies use simple material balance, Rate Transient Analysis (RTA), and decline curve analysis to characterize the drainage area and to predict the ultimate recoveries from wells. Lee and Sidle (2010) provide a detailed critique of different types of methods used to forecast production in unconventional reservoirs. In the last few years, advances in academia and the industry have led to advances in multiphase RTA (Eker et al., 2014; Qanbari and Clarkson, 2016; Bowie & Ewert, 2020). While these methods are simple and easy to use, it should be noted that decline curve analysis is not pressure normalized and RTA models are production data dependent. (They cannot predict changes in production because of changes in hydraulic fracture). In light of this, a hydraulic fracture model that is independent of production data and can be fed seamlessly into a reservoir simulation model is necessary.

Hydraulic fracture modeling has come a long way since the development of the Khristianovich-Geertsma-de Klerk (KDG) and the Perkins-Kern-Nordgren (PKN) 2D models in the 1960s (Khristianovitch and Zheltov, 1955; Geertsma and Klerk, 1969; Perkins and Kern, 1961). The fully 3D fracture model used in this paper can now simulate fracture geometry and propagation in the vertical and horizontal direction in multilayered formations, incorporating rock mechanics to account for stress changes, fracture height growth, and stress shadowing (Barree, 1983, & Barree & Miskimins, 2003). In this model, the influence of natural fractures can be introduced explicitly with various leak-off parameters. While this model is sophisticated and can handle a lot of complexity, the model still needs to be calibrated to DFIT and microseismic data to increase the fidelity of the model.

The presence and magnitude of natural fractures can be detected by looking for the Pressure Dependent leak-off (PDL) in the DFIT analysis, as demonstrated by Barree et al. (2007). Other parameters from the DFIT analysis will also be used for the calibration of the hydraulic fracture. Additionally, events from microseismic data can be used to calibrate the overall fracture geometry, as demonstrated by Cipolla et al. (2011).

The geomechanics of the reservoir is also critical for designing and understanding hydraulic fracturing because they provide rock strength and stress information, which directly influence how fractures initiate, propagate, and grow. Of utmost importance is the minimum horizontal stress and reservoir pressure, which can be calibrated to DFIT.

Pressure interference tests involve flowing one well (in a pad) and monitoring the pressure changes in shut-in offset wells. The timing and magnitude of the response indicate how connected the wells are. Chu et al. (2018) introduced the Chow Pressure Group (CPG) to quantify the magnitude of pressure interference. The table below shows that a CPG value of 0 signifies no pressure interference, less than 0.5 signifies weak interference, 0.5 to 0.8 is moderate, and greater than 0.8 signals strong interference.

Table 1—Pressure interference catergory from Chu et al.

CPG value	Level of Interference
0	No pressure interference
0-0.5	Weak
0.5-0.8	Moderate
0.8-1	Strong

The pressure interference is a valuable tool to calculate both the hydraulic fracture length and reservoir pressure response in a dynamic reservoir simulator.

Once the hydraulic fracture model has been calibrated, the resulting fracture conductivity and geometry can be imported into a dynamic reservoir simulator for production history and sensitivity studies. It is well known that the process of history matching production data can be non-unique, therefore the pressure output from the simulator was compared to the result of the inteference test.

Laid out above is about 3 to 5 domains in petroleum engineering and geology that must be understood in order to properly predict gas and oil from an unconventional reservoir. Obviously that a comprehensive workflow is needed that can capture and connect all features and at the same time honor the underlining physics as rigorous as possible. The aim of the paper is to encourage a more holistic approach to the production modelling of unconventional reservoirs. The workflow proposed in this paper is highlighted in fig 1. It is the authors' opinion that if each step of the process is done correctly, the final answer will be representative of the reservoir production mechanism.

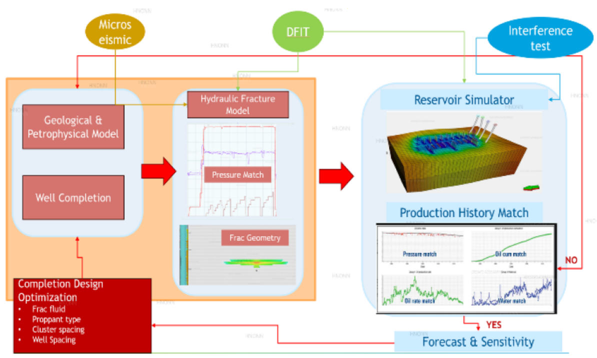

Figure 1—A holistic workflow to modelling production from unconventional reservoirs.

PAD Overview

This study was implemented on two adjacent 2-well-pad configurations with slickwater completion as shown in figure 2 below. The wells are equally spaced. Well 3 is landed in a shallower formation compared to Well 1, 2 and 3, and reservoir fluid is a retrograde condensate gas.

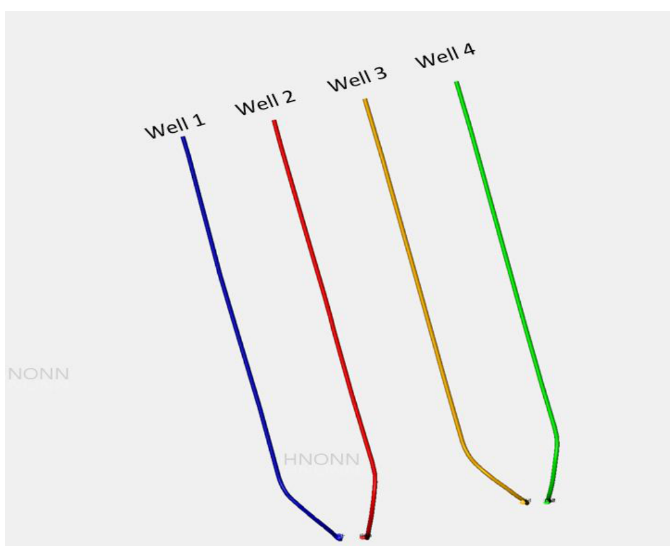

Figure 2—4 Wells in the area of study

Microseismic was performed in well 1 and in well 3, and pressure interference was performed on all the wells as demonstrated in the pressure interference section. A vertical pilot well with petrophysis and geomechanical logs was drilled in the vertical section of well 1, and a DFIT analysis was performed on the toe stage of well 2.

DFIT Analysis

The workflow starts with calibrating hydraulic fracture model with the result of the DFIT analysis as shown in table 2.

Table 2—Important parameters from DFIT to calibrate hydraulic fractures

Paramater	Definition	Uses
Closure pressure	Pressure at which hydraulic fracture closes after pumping.	Calibration of minimum horizontal stress. Critical for frac geometry
Reservoir pressure	Fluid pressure within the pore space of the reservoir.	Used in the calculation of effective stress.
Reservoir permeability	Ability for rock to transmit fluid.	Affects leakoff from hydraulic fractures and used to calibrate reservoir SRV permeability
Process Zone Stress	Elevated stress region near the tip of the frac. Includes mircocracks and inelastic deformation.	Calibration of frac net pressure and frac height containment
Fissure opening pressure	Pressure that natural fracture opens and leakoff into NF starts.	Affects fracture geometry (mainly frac length)
Pressure dependent leakoff	Rate of leakoff into natural fractures.	Affects fracture geometry (mainly frac length)

The purpose of the DFIT is to estimate reservoir rock closure pressure for the calibration of the minimum horizontal stress, to estimate the initial reservoir pressure for the calibration of the pore pressure to be used in the geomechanical model and reservoir simulation model, for general reservoir diagnostic (presence of natural fractures), reservoir permeability, leakoff, and for net pressure calibration for the hydraulic fracture model.

50 barrels of 2% KCl were pumped, and the fall-off pressure was recorded to ensure sufficient time to observe reservoir characteristics, following the recommendation of Barree et al. (2014). The pressure fall-off plot in Figure 3 shows a clear breakdown, while the G-function plot in Figure 4 shows a pressure-dependent leak-off. The process zone stress is the difference between the instantaneous shut-in pressure (ISIP) and the closure pressure. The closure pressure is directly estimated from the G-function plot, as shown in Figure 4. The convex-down shape of the superposition derivative in Figure 4 signifies a pressure-dependent leak-off, indicating that the fluid is leaking off from the fracture at a rate higher than normal fluid leak-off. The physical explanation for this is that there may be natural fractures present in the formation, causing the excessive leak-off. Hence, this diagnostic can be used to identify the presence of natural fractures in the formation. The rate pressure-dependent leak-off) at which the leak-off occurs into the natural fractures is estimated from Figure 5. Lastly, the reservoir pressure is directly estimated from the after-closure linear flowplot.

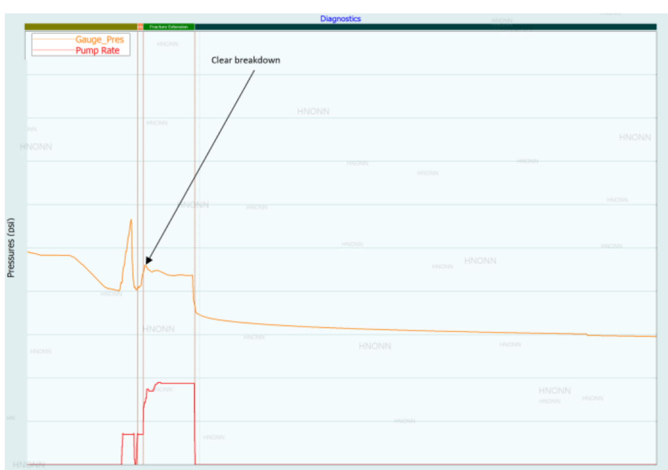

Figure 3—DFIT test and pressure falloff history

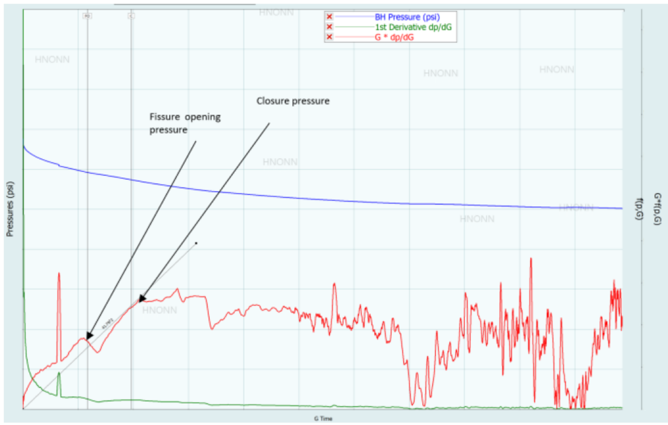

Figure 4—G-Function Analysis

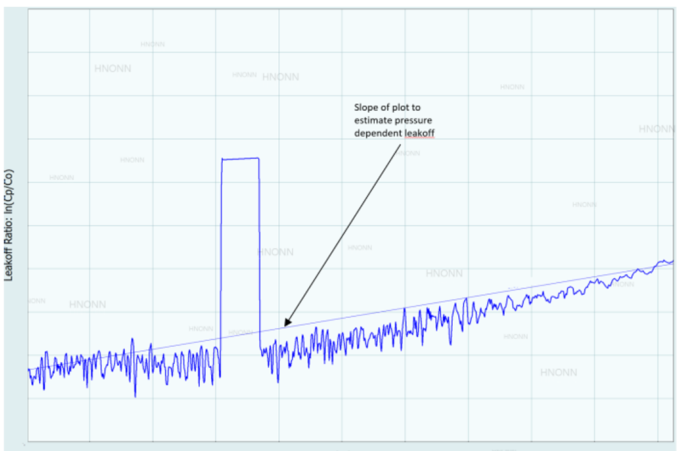

Figure 5—Pressure dependent analysis plot (PDL)

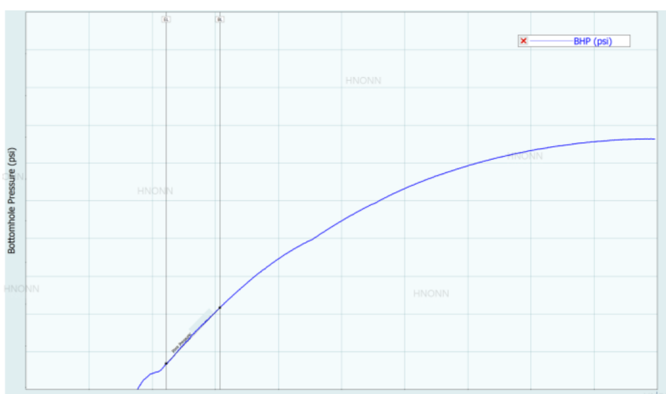

Figure 6—After Closure Linear Flow plot

Hydraulic Fracture Pressure Matching

The next step in the process is to calibrate the hydraulic fracture model and perform a pressure match, where the field treating pressure is matched against the model treating pressure. The DFIT parameters in Table 2 are used to calibrate the hydraulic fracture model, and during the pressure matching process, the only parameters adjusted were the frictional parameters (casing friction and perforation friction). Figure 7 shows the result of the pressure match for Stage 1. The field treating pressure is shown in purple, while the model treating pressure is shown in black. The red curve and yellow curve represent the slurry rate and proppant concentration, respectively. In the hydraulic fracture pressure matching process, it is critically important to match the pressure decline after ISIP, because this portion provides vital information about the fracture closure behavior, leakoff characteristics, and formation pressure properties. Figure 7 shows that there is an excellent match between the model pressure and the field pressure after ISIP.

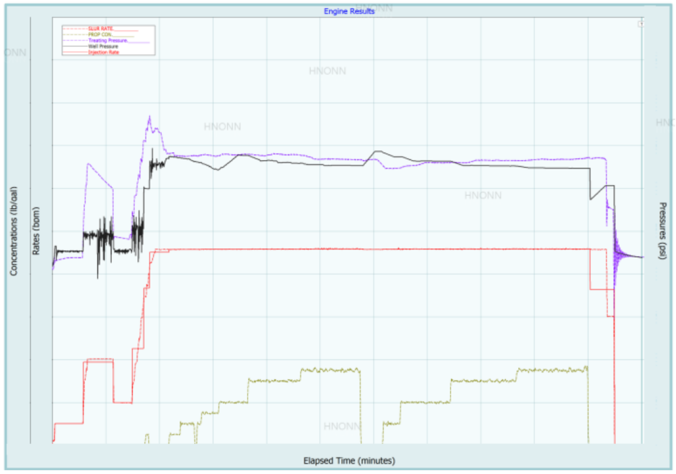

Figure 7—Result of hydraulic fracture pressure matching

Figure 8 shows the result of the frac geometry for 1 cluster for one stage. The result is as expected, and the frac geometry is strongly dictated by the geomechanical properties of the reservoir. The intensity of the color represents the conductivity of the hydraulic fracture.

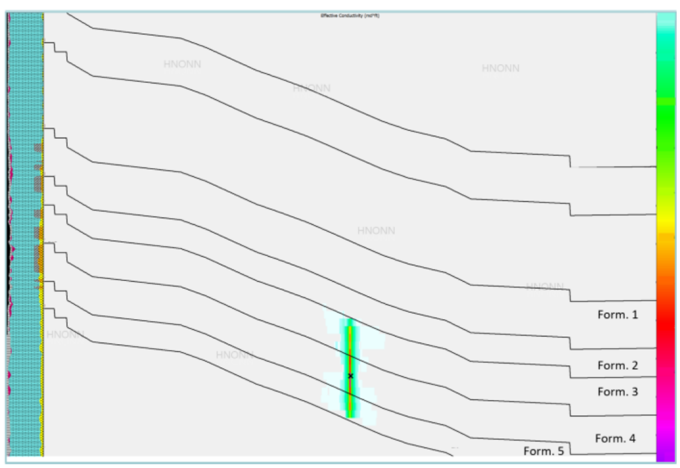

Figure 8—Result of frac geometry & conductivity distribution for 1 frac in 1 stage.

The fracture geometry for the stage that was modeled was then replicated to be used for the other stages in the well. The micro seismic events were then overlayed on the hydraulic fracture as shown in figure 9 and figure 10 respectively. The line represents the hydraulic fracture (not conductive) length and height. This frac model is refered to as Frac Model A. In this model, the hydraulic fracture length matches reasonably well with the microseismic length for Well 1 and Well 2. On the other hand, the the hydraulic fracture height only matches with the microseismic heighgt for Well 1 and not for Well 3.

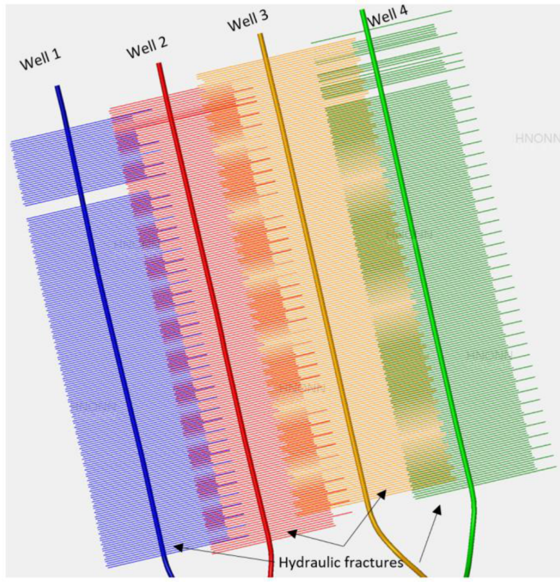

Figure 9a—Hydraulic fracture length for the four wells represented by polygons.

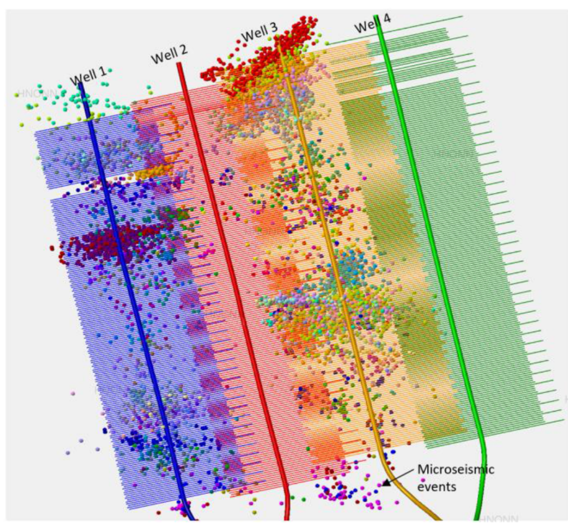

Figure 9b—Microseismic events overlayed with hydraulic fractures.

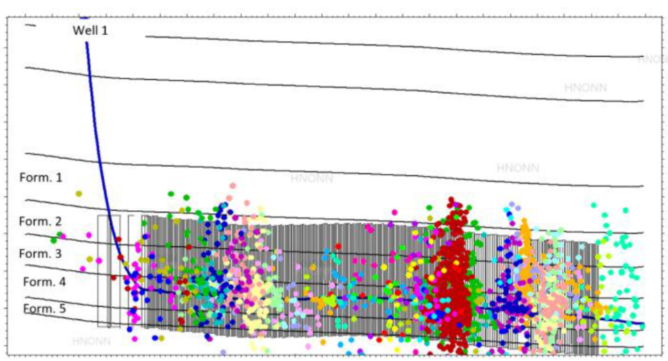

Figure 9c—Microseismic events for Well 1 overlayed with the hydraulic fractures.

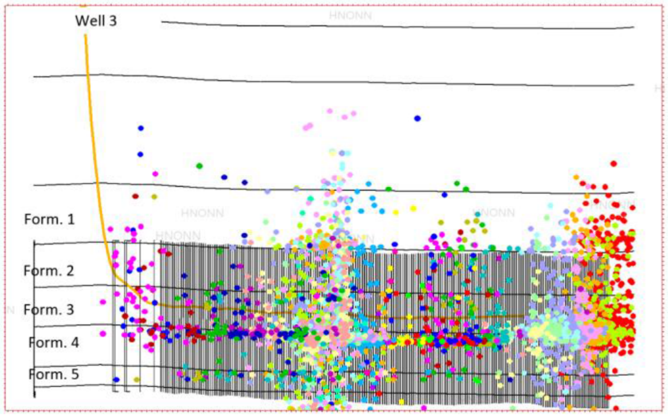

Figure 9d—Microseismic events for Well 3 overlayed with the hydraulic fractures.

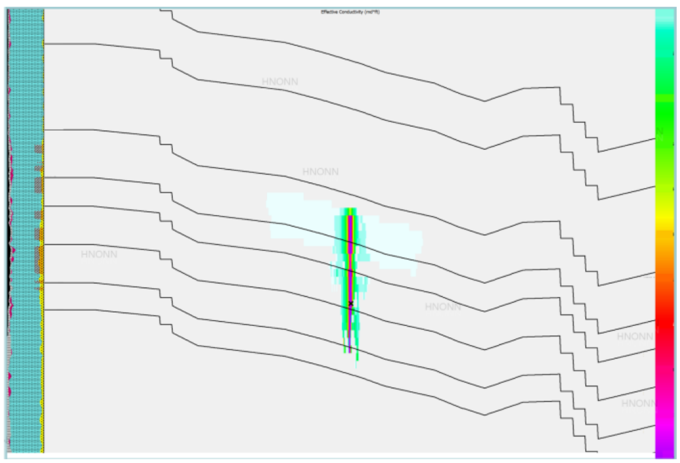

Figure 10—Result of frac geometry & conductivity distribution for 1 frac in 1 stage for Model B.

In Figure 9, we notice that there are quiet events for some stages when fraced. Also, for Well 3, we see a significantly higher number of events in Formation 1 compared to Well 1. One explanation for the quiet stages is that the stress shadow from previous stages increases the compressive stress around adjacent untreated intervals, which raises the minimum horizontal stress locally, making shear or tensile failure less likely. This can cause new fracturing events to occur in an aseismic manner with low energy release, possibly below the detection threshold of the monitoring array. Another explanation for the quiet events is that the lack of pre-existing natural fractures (NF) in that area reduces the likelihood of shear reactivation, releasing very low energy. It is possible that the microseismic events seen at the toe and the middle of the well for Well 3 are solely due to the reactivation of natural fractures.

In order to rigorously match the microseismic data, another frac model was developed, referred to as Frac Model B. In this model, the leakoff parameters were changed for different stages to match the microseismic events. Figure 11 shows the frac geometry and conductivity distribution of the frac geometry for one cluster and one stage for Model B. The result is dictated by the stress shadow intensity, the process zone stress and leakoff parameters.

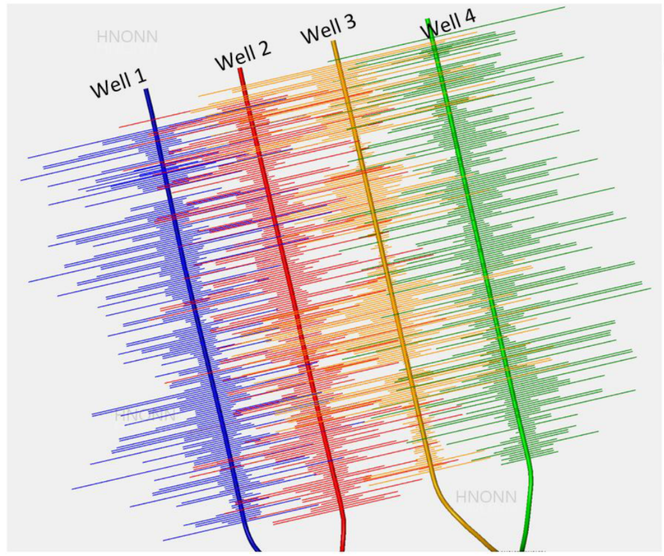

Figure 11a—Hydraulic fracture length for the four wells represented by polygons for Model B. The frac geometry is dictated by stress shadow intensity and leakoff parameters.

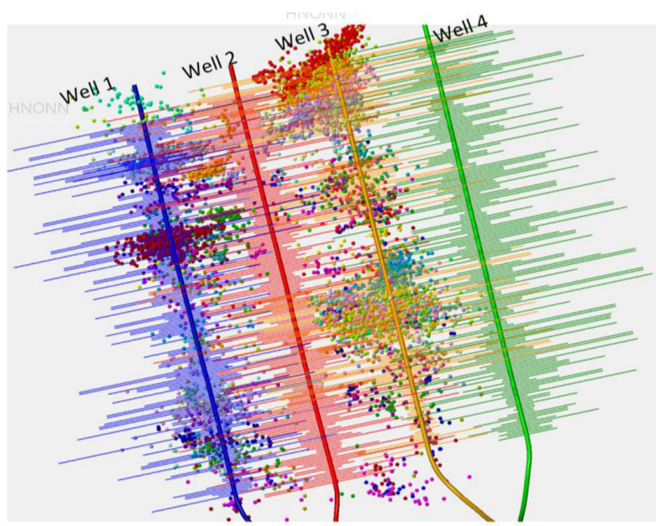

Figure 11b—Microseismic events overlayed with hydraulic fractures for Model B.

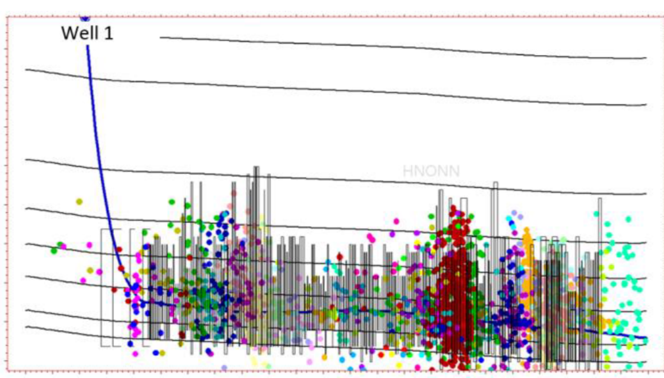

Figure 11c—Microseismic events overlayed with hydraulic fractures for Well 1.

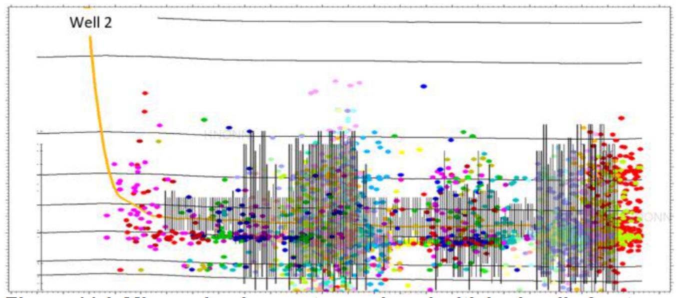

Figure 11d—Microseismic events overlayed with hydraulic fractures for Well 3.

In summary, two hydraulic fracture models were developed, Model A, where one stage was rigorously calibrated to the DFIT and the replicated to represent the other stages of the well, and Model B, where each individual stage was rigously matched to microseismic.

Intereference Test

The interference test was designed to maximize the information obtained from the 4 wells on the pad. All wells were initially flowing, and at a particular time, they were all shut in to allow the pressure in the reservoir to re-equilibrate for a period of time. After this, the test was conducted as described in Figures 12 and 13. Well 2 served as the monitoring well during the interference test and was never flowed. Well 3 was opened first to test the interference between Well 3 and Well 2. Well 4 was opened next, and Well 1 was opened last.

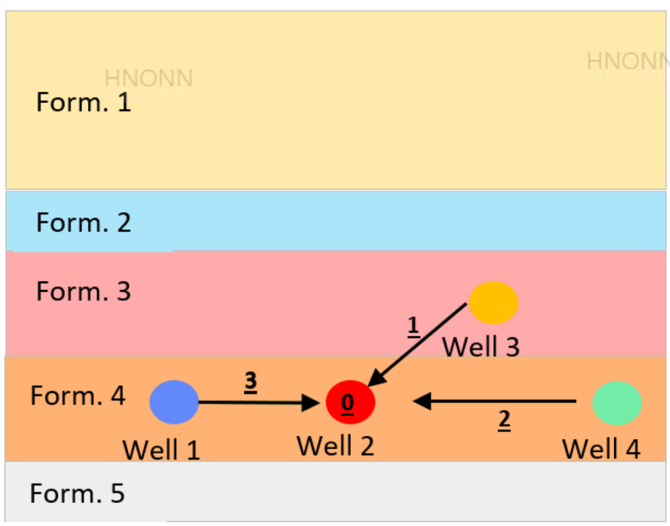

Figure 12—Cartoon of the interefence test. Well 2 is the monitoring well. The underscore numbers represent the order of the well openning

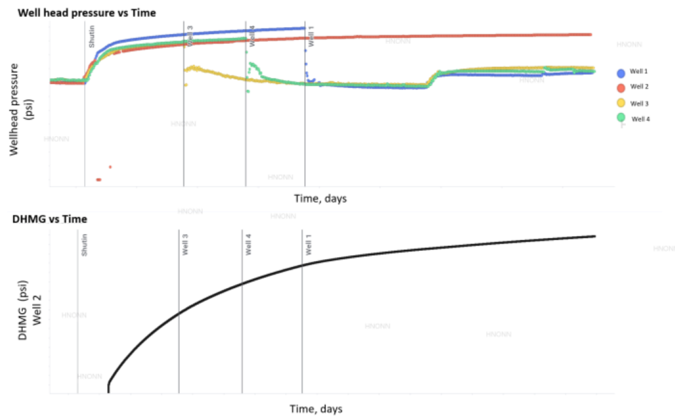

Figure 13—The down hole gauge pressure respone of Well 2 when Well 3, Well 1 and Well 4 are opened

When Well 1, Well 3 and Well 4 are opened, no observable inflection is noticed on the downhole memory gauge of Well 2. To verify there are no inflections and there are no well to well pressure inteference, the CPG method analysis developed by Chu et al. (2018) was used.

In Figure 14, the CPG is shown in green, the pressure difference (DP) in blue, and the derivative of the pressure difference in red. The CPG analysis of Figure 14 indicates that there is no pressure interference, as the value is close to 0. CPG pressure interference analysis was also conducted for the other two scenarios:

2: Pressure interference of Well 4 on Well 2

3: Pressure interference of Well 1 on Well 2

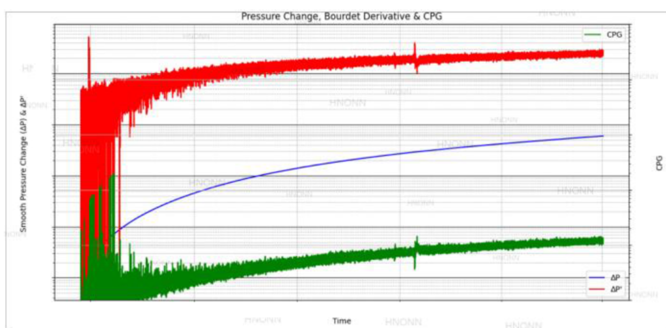

Figure 14—Pressure interference analysis of opening Well 3 on Well 2

All the CPG analyses show negligible pressure interference. Since the pressure interference test shows no pressure interference among the wells, it is more likely that Frac Model A is a better representation of the hydraulic fractures than Frac Model B. To test this, the two frac models were imported into a dynamic reservoir simulator.

Reservoir Simulation

Before proceeding to well spacing optimization, the fidelity of the two frac models was tested in a reservoir simulator where global cells around the frac were divided into smaller regions using a local grid refinement (LGR) method. Hydraulic fracture geometries were imported into the reservoir simulator using a conductivity cut-off value of 0.01 md-ft. Figure 14a shows a total of 4 regions: the fracture in green, the first stimulated reservoir volume (SRV1) in light green, the second stimulated reservoir volume (SRV2) in red, and the matrix in blue.

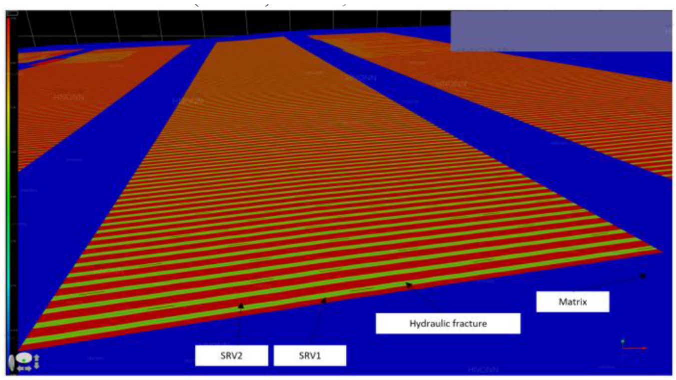

Figure 14a—Different regions around hydraulic fracture to assist in reservoir simulation history match.

The reservoir pressure around the well can be elevated to account for the pressure supercharge effect after fracing a well. The matrix, hydraulic fracture, SRV1, and SRV2 permeabilities, as well as the pressure-dependent permeability and porosity, and the relative permeability table, are all history-match parameters. In this study, a constant compaction curve and relative permeability were used for all four wells in order to avoid overfitting and to make the model usable in other areas of the field. A modified black oil table was used for the fluid model. To match the produced frac water, the water saturation in and around the hydraulic fractures was changed.

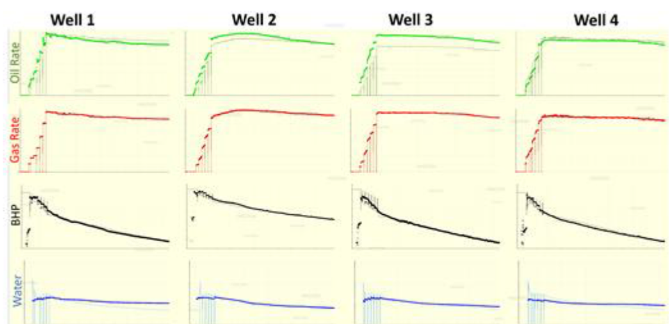

Figure 15—History Match Result for frac Model A.

The history match was controlled by automatically matching the gas rate, while the reservoir properties and frac conductivity were changed to match the bottom hole pressure (BHP), oil rate, and water rate. The oil match is excellent for Wells 1, 2, and 4. The reason the oil match might not be as good for Well 3 is that it is landed in a shallower formation, and the fluid properties might be slightly different. The bottom hole match is also excellent.

Figure 16 shows 3 main pressure regimes in the reservoir. The original virgin reservoir pressure, an elevated reservoir pressure area to simulate the pressure supercharge effect after fracing, and the depleted pressure area around the frac and the well. For all four wells, the depleted pressure areas around the fracs do not touch. This signifies that the wells are not in communication, as also confirmed by the pressure interference test.

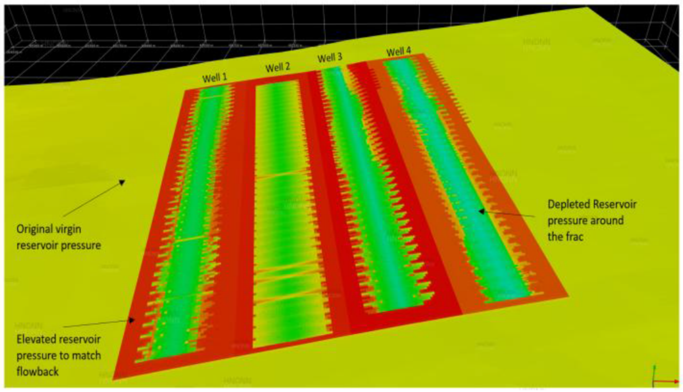

Figure 16—Pressure depletion profile for Frac Model A.

Figure 17 shows the pressure depletion profile for Frac Model B. As expected, we see an irregular pressure depletion profile, and the depletion profile follows the hydraulic fracture geometry. We also observe some pressure interference since the depletion profile of the wells overlap.

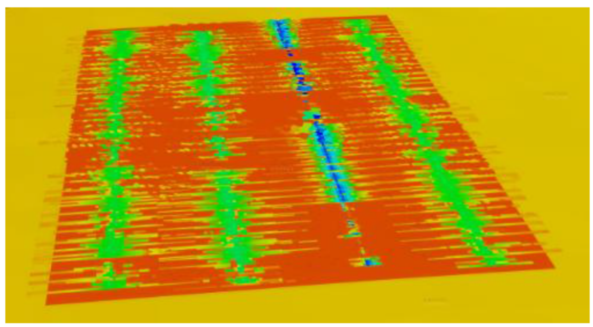

Figure 17—Pressure depletion profile for Frac Model B

Creating the history match in Figure 18 was challenging due to the interaction of reservoir properties across different regions of each well. On average, the (xf) for Well 3 is the smallest. To match production, the conductivity and SRV permeability must be increased, resulting in greater depletion around the well.

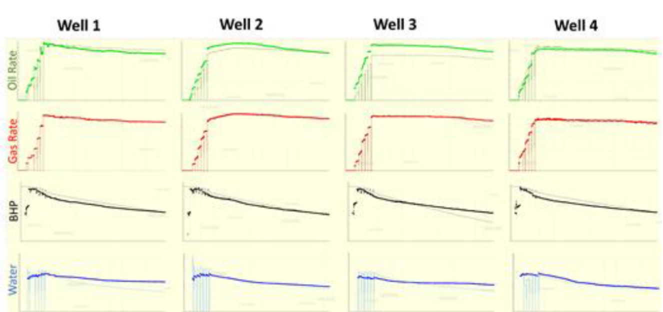

Figure 18—Pressure depletion profile for Frac Model B

Since no pressure interference was observed from the pressure interference test, and the simulation for Frac Model B indicates interference, combined with the fact that microseismic events generally exaggerate hydraulic fracture geometry (as demonstrated by Warpinksi & Wolhart, 2016), it is more likely that Frac Model A is correct compared to Frac Model B.

Results of Rate Transient Analysis

After a rigorous history match in the Reservoir Simulation Domain, a Rate Transient Analysis (RTA) was performed on the production data for all the wells. Figure 19, as an example, shows the log log analysis for Well 1. The Log-Log plot in Figure 19 shows two flow regimes: bilinear flow and linear flow. The xf estimated from the RTA is within the range of the xf estimated from the hydraulic fracture simulation for Model A. Based on this fact, it was concluded that Model A is likely more accurate than Model B

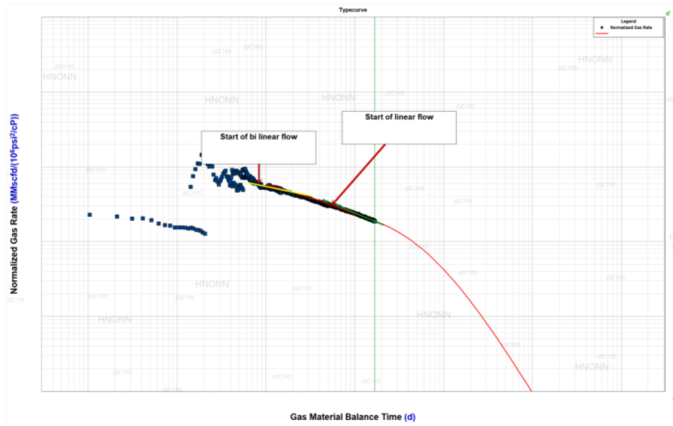

Figure 19—Log-Log Plot for Well 1

Conclusion

A holistic, physics-based workflow has been developed and demonstrated to improve the reliability of production forecasts and well spacing optimization in unconventional reservoirs. The methodology integrates multiple disciplines—DFIT calibration, microseismic interpretation, hydraulic fracture modeling, pressure interference analysis, and dynamic reservoir simulation—to construct a fully calibrated model that honors both reservoir and fracture characteristics.

The workflow begins with rigorous DFIT interpretation to constrain key geomechanical parameters such as minimum horizontal stress and reservoir pressure, which are essential inputs for fracture modeling. The hydraulic fracture geometry is then validated using microseismic data, enhancing confidence in fracture propagation and complexity. Pressure interference testing, analyzed through the Chow Pressure Group method, provides an independent measure of fracture connectivity and is used to verify the reservoir simulation outputs. This multi-domain calibration substantially reduces uncertainty associated with model predictions and production history matching.

The approach enables simulation and completion engineers to make technically sound decisions regarding fracture design, completion strategies, and well spacing in similar geologic and petrophysical settings. By incorporating high-fidelity, field-calibrated inputs across multiple data sources, this workflow advances reservoir understanding and improves the predictability of unconventional field development.

Nomenclature

Δp = Pressure Differential (psi)
q_w = Water Injection Rate (STB/D)
P_{wfi} = Bottomhole Pressure (psi)
P_e = Reservoir Pressure (psi)
T = Transmissibility (D.ft or mD.ft)
k = Permeability (Darcy D or millidarcy mD)
h = Thickness (ft)

F_{rr} = Rresidual Resistance Factor
RTA = Rate Transient Analysis
Dfit = Diagnostic Fracture Injection Test
CPG = Chow Pressure Group Analysis
PDL = Pressure Dependent Analysis Plot
NF = Natural Fractures
SRV = Stimulated Rock Volume
xf = Fracture Half Length
hf = Fracture Half hight

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